

ChannelShuffle#

<https://pytorch.org/docs/stable/generated/torch.nn.ChannelShuffle.html>

Description:

Divides and rearranges the channels in a tensor. This operation divides the channels in a tensor of shape $(N, C, *)$ into g groups as $(N, \frac{C}{g}, *)$ and shuffles them, while retaining the original tensor shape in the final output. `groups (int)` – number of groups to divide channels in. Return the extra representation of the module.

Function Signature

```
class torch.nn.ChannelShuffle(groups)[source]#
```

Parameters

```
extra_repr()[source]#
```

Return type

```
forward(input)[source]#
```

Returns:

Tensor

Code Examples

```
>>> channel_shuffle = nn.ChannelShuffle(2)
>>> input = torch.arange(1, 17, dtype=torch.float32).view(1, 4,
2, 2)
>>> input
tensor([[[[ 1.,  2.],
           [ 3.,  4.]],
        [[ 5.,  6.],
           [ 7.,  8.]],
        [[ 9., 10.],
           [11., 12.]],
        [[13., 14.],
           [15., 16.]]]]])
>>> output = channel_shuffle(input)
>>> output
tensor([[[[ 1.,  2.],
           [ 3.,  4.]],
        [[ 9., 10.],
           [11., 12.]],
        [[ 5.,  6.],
           [ 7.,  8.]],
        [[13., 14.],
           [15., 16.]]]]])
```

Detailed Documentation

Documentation

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Return the extra representation of the module.

Runs the forward pass.

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